

⚡ OFFICIAL PRODUCT WHITEPAPER & USER MANUAL

Minomeet AI

Autonomous On-Device & Cloud Meeting Intelligence, Real-Time Speech
Diarization, and Executive Minutes of Meeting (MOM) Platform



Designed for Modern Engineering Teams, Executives, Product Managers, and Privacy-Conscious Organizations.

VERSION

v1.2.0 Production Release

CORE ENGINE

Dual-Audio STT + Frontier AI LLMs

AUTHOR & MAINTAINER

@shareefmx
(github.com/shareefmx)



Table of Contents

Section	Title & Topic	Target Page
Chapter 1	The Meeting Dilemma & Problem Statement	Page 3
Chapter 2	Core Solution & System Architecture	Page 4
Chapter 3	Screen-by-Screen Feature Guide & UI Walkthrough	Page 5–7
Chapter 4	Multi-Model AI Control Panel & Provider Setup	Page 8
Chapter 5	1-Click Follow-Up Emails & Universal Export	Page 9
Chapter 6	Step-by-Step Installation & Operations Manual	Page 10
Chapter 7	Privacy, Security Governance & Troubleshooting	Page 11



Chapter 1: The Problem Statement

In today's hybrid work environment, professionals spend an average of **21.5 hours per week** across Google Meet, Zoom, Microsoft Teams, and Slack Huddles. Despite this massive investment of time, post-meeting execution remains severely fractured due to four critical bottlenecks:

1. Note-Taking Fatigue & Divided Focus

Participants are constantly distracted trying to jot down conversation fragments instead of actively contributing to strategic discussions. Key verbal subtleties and context are lost.

2. "Meeting Amnesia" & Lost Deliverables

Decisions made verbally during a 45-minute sprint planning call often vanish. Action items lack explicit ownership and due dates, leading to redundant follow-up calls.

3. Time Waste on Manual MOM Summaries

Drafting executive summaries and emailing team recaps takes 30 to 60 minutes of tedious administrative effort after every single meeting.

4. Invasive Cloud Bots & Privacy Leaks

Traditional AI note-takers require third-party bots to join calls as uninvited attendees, storing confidential intellectual property on opaque external cloud servers.

The Core Objective of Minomeet AI: Provide an autonomous, 100% private, on-device and cloud-flexible meeting intelligence platform that captures conversations directly from hardware with zero invasive bots, synthesizing raw speech into executive-ready deliverables in under 5 seconds.

Chapter 2: Core Solution & System Architecture

Minomeet AI employs a **4-Stage End-to-End Intelligence Pipeline** that isolates audio ingestion, live transcription, neural synthesis, and structured storage directly on the local machine:

SYSTEM ARCHITECTURE TOPOLOGY

100% ON-DEVICE & DIRECT FRONTIER AI

[AUDIO INGESTION] → [REAL-TIME STT] → [NEURAL SYNTHESIS] → [STRUCTURED OUTPUT] •
Microphone • Web Speech API • Google Gemini • Executive Summary • System Audio • Neural Whisper •
OpenAI GPT-4o • Key Decisions (3) • Video Conference • Speaker Diarize • Claude 3.7 Sonnet • Action
Items Matrix • Audio Files (.mp3) • Millisecond Stamps • Groq LPU / Ollama • 1-Click Email Recap

Key Architectural Principles

Layer	Engine / Tech Stack	Value Proposition
Dual Audio Capture	WebRTC AudioContext + System Loopback	Captures user mic and remote attendees simultaneously with real-time waveform equalization.
Speech-to-Text (STT)	Browser Neural STT + Whisper Transformer	Converts live voice to text with speaker labeling and continuous time-aligned transcripts.
Dynamic AI Resolver	Multi-Provider Client/Server Orchestrator	Single source of truth routing prompts to Google, OpenAI, Claude, Groq, or local Ollama.
Local Data Store	LowDB Atomic JSON Archive	100% private local database stored at <code>server/data/db.json</code> . Zero external telemetry.

 **Privacy Guarantee:** Minomeet never transmits raw audio files to central servers. When cloud AI synthesis is enabled, only anonymized textual transcripts are sent directly to your chosen provider via your private API key.

3.1. Main Dashboard & Quick Action Hub (HomeScreen.tsx)

The Home Screen provides a centralized control hub displaying real-time productivity metrics, meeting archives, and the prominent AI Model Setup Banner:

```
UI SNAPSHOT — HOME SCREEN (DASHBOARD)                                VIEWPORT: DESKTOP 1080P

+-----+ |
< Minomeet AI [ Search Meetings... ] [ + Start Live Recording ] | +-----+
+-----+ | 🔑 [ ALERT BANNER ] AI Model Not
Configured – First-Time Setup Required [ Configure AI Key ] | +-----+
+-----+ | 🌟 Welcome to Minomeet AI – Next-Gen
Autonomous Meeting Intelligence | | 📁 Start Live Recording ] [ 📄 Import Audio File ] [ 💬 Ask
Your Meetings ] | +-----+
+-----+ | ┌──────────┴──────────┐
└──────────┘ | | TOTAL MEETINGS: 12 | | MOM CONVERTED: 11 | | CONVERSION RATE:
92% | | | Archived on Device | | Structured Intelligence | | Zero Effort Minutes | | 
└──────────┘ └──────────┘ └──────────┘ | +-----+
+-----+
```

Metric Indicators

- **Total Meetings:** Real-time count of all sessions transcribed.
- **MOM Converted:** Percentage of sessions with generated executive summaries.
- **Recent History:** Instant 1-click access to past meeting notes and audio logs.

Action Buttons

- **Start Live Recording:** Launches the studio with dual-audio capture.
- **Import Audio:** Drag-and-drop `.mp3` , `.wav` , `.m4a` , `.webm` .
- **Ask Your Meetings:** Launches the multi-meeting semantic search chatbot.

3.2. Live Studio & Waveform Engine (RecordingScreen.tsx)

The Recording Screen manages live audio streams with dual-channel volume metering, real-time speech diarization, and dynamic waveform visualization:

UI SNAPSHOT — RECORDING STUDIO & LIVE TRANSCRIPTION

STATUS: LIVE STREAMING (00:14:32)

LIVE RECORDING [00:14:32] | Source: Mixed (Mic + System) [ Pause] [ Stop MOM] |

Dynamic Dual-Channel Audio Waveform (Frequency Equalizer + Gain Control) → |

LIVE SCROLLING

TRANSCRIPT WITH SPEAKER DIARIZATION: | | | [14:10] Speaker 1 (You): "Let's review the API key architecture for the v1.2.0 release." | [14:18] Speaker 2 (Alex): "Agreed. Google Gemini and OpenAI are fully connected." | [14:26] Speaker 1 (You): "Sarah, please finalize the action items matrix by tomorrow at 5 PM." |

Chapter 3 (Continued): Meeting Intelligence & MOM Notes

3.3. Executive Minutes of Meeting (NotesScreen.tsx)

Once recording stops, Minomeet automatically extracts conversations into clean, structured sections formatted for executive distribution:

```
UI SNAPSHOT — EXECUTIVE MINUTES OF MEETING & ACTION MATRIX                                MODEL: GEMINI 2.5 FLASH

+-----+ |
📄 Sprint Planning & Product Launch v1.2.0 Date: Aug 28, 2026 • 45 Mins • 5 Attendees | | [ 🔄 Re-
Synthesize ] [ ✉ Email Recap ] [ 📄 Export Markdown ] [ 🖨 Print PDF ] [ 🗑 Delete ] | +-----+
+-----+ | 📌 1.
EXECUTIVE SUMMARY | | The team reviewed final deployment milestones for v1.2.0, standardizing AI
provider APIs | | and verified zero telemetry on-device storage. Launch scheduled for Friday
morning. | +-----+
+-----+ | 🗨 2. KEY DECISIONS MADE | | ✓ Decision 1: Standardize on Google Gemini 2.5 Flash as
default high-speed cloud LLM. | | ✓ Decision 2: Remove built-in model weights to keep app package
lightweight and agile. | | ✓ Decision 3: Enforce strict local JSON database storage for corporate
security. | +-----+
+-----+ | ✅ 3. ACTION ITEMS MATRIX (DELIVERABLES) | | [x] @Sarah – Finalize documentation and
release notes | Due: Tomorrow, 5:00 PM | | [ ] @Alex – Run automated regression tests across all
providers | Due: Friday, 10:00 AM | | [ ] @Dev – Push tag v1.2.0 to GitHub release repository | Due:
Friday, 12:00 PM | +-----+
+-----+
```

3.4. Semantic "Ask Your Meetings" AI Assistant (AskMeetingsModal.tsx)

Enables users to query past discussions across months of archives using natural language with instant citations:

```
UI SNAPSHOT — SEMANTIC Q&A CHATBOT                                                    HISTORICAL ARCHIVE SEARCH

+-----+ |
💬 Ask Your Meetings AI Assistant | +-----+
+-----+ | User: "What did we decide about the Q3 budget allocation in
yesterday's call?" | | | Minomeet AI: | | "In yesterday's Sprint Planning meeting at timestamp
[18:42], Alex approved a total budget | | of $15,000 allocated towards server infrastructure, with
$5,000 reserved for Groq LPU APIs." | | | Source Reference: [Meeting #12: Sprint Planning →
Timestamp 18:42] | +-----+
+-----+ | [ Enter question about past meetings... ] [ 📤 Send ] | +-----+
+-----+
```

Chapter 4: Multi-Model AI Control Panel & Setup

Minomeet enforces **Dynamic Model Resolution** as the single source of truth across all meeting synthesis tasks:

AI MODEL RESOLUTION LOGIC

STATUS: DYNAMIC AUTO-FALLBACK

Settings → Active Provider → Model Selection → Test Connection → Save → Synthesize

Supported AI Providers Comparison

Provider	Recommended Models	Connection Endpoint	Best For
Google Gemini	gemini-2.5-flash , gemini-2.5-pro	Google AI Studio API	Ultra-fast, massive context window (1M tokens), high reasoning.
OpenAI	gpt-4o , gpt-4o-mini , o3-mini	api.openai.com/v1	Industry standard for structured JSON schema extraction.
Anthropic Claude	claude-3-7-sonnet , claude-3-5-sonnet	api.anthropic.com/v1	Superior prose, natural executive tone, deep synthesis.
Groq	llama-3.3-70b-versatile	api.groq.com/openai/v1	Ultra-low latency (500+ tokens/sec) on LPU inference hardware.
Ollama (Local)	llama3.3:70b , qwen2.5:7b	http://127.0.0.1:11434	100% on-device, offline, zero cloud calls.
Custom / vLLM	OpenAI-compatible endpoints	Custom Base URL	Enterprise internal LAN LLM clusters and LM Studio.

How to Configure Your API Key (3 Simple Steps)

1. Open **Settings** from the left sidebar and click on the **AI Model** tab.
2. Select your preferred AI provider (e.g., **Google Gemini**) and paste your secret API key.
3. Click **"Test Connection"** to verify real-time connectivity, then click **"Save Global Configuration"**.

✉ Chapter 5: 1-Click Follow-Up Emails & Universal Export

5.1. 1-Click Tailored Email Generator (FollowUpEmailModal.tsx)

Turn Minutes of Meeting into ready-to-send emails with 3 distinct organizational tones:



Professional Tone

Comprehensive executive overview, formal attendee greetings, full decision context, and structured deliverable tables.



Concise Bulleted

Scannable, high-velocity bullet points highlighting only the top 3 takeaways and critical immediate deadlines.



Action-Oriented

Task-first breakdown grouping deliverables by team member (@Sarah, @Alex) with bold due dates.

UI SNAPSHOT — FOLLOW-UP EMAIL DRAFT GENERATOR

ACTION: 1-CLICK CLIPBOARD COPY

Subject: Follow-up & Executive Summary: Sprint Planning v1.2.0 [Aug 28, 2026] Hi Team, Thank you for attending today's sprint review. Here is a summary of key decisions and next steps: 📌 Executive Summary: • Finalized v1.2.0 deployment milestones and verified local JSON data security. • Launch scheduled for Friday morning with Gemini 2.5 Flash as default engine. 🎯 Action Deliverables: 1. @Sarah – Complete documentation and release notes (Due: Tomorrow 5 PM) 2. @Alex – Execute end-to-end regression tests (Due: Friday 10 AM) Best regards, Minomeet AI Executive Assistant

5.2. Multi-Format Universal Exporter

Export Format	File Type / Extension	Target Use Case
Markdown (.md)	Raw Github-flavored markdown	Obsidian, Notion, GitHub wikis, developer documentation.
Plain Text (.txt)	UTF-8 Plain Text	Slack messages, email pastes, raw text archives.
Print to PDF	High-resolution vector PDF	Executive distribution, client reports, board decks.

Chapter 6: Installation & Operations Manual

Prerequisites

- **Node.js:** `v18.0.0` or higher (`v20+` LTS strongly recommended)
- **npm / pnpm / yarn** package manager
- **Python 3.10+** (Optional: for local neural speech transcription)
- **Browser:** Chrome, Edge, Brave, or Safari with microphone permissions enabled

Step-by-Step Installation Commands

```
# 1. Clone the repository from GitHub
git clone https://github.com/shareefmx/minomeet.git
cd minomeet

# 2. Install root, client, and server dependencies
npm install

# 3. (Optional) Install Python packages for on-device neural transcription
pip install -r requirements.txt

# 4. Start the development server
npm run dev
```

- **Client Web App:** `http://localhost:5173`
- **Server Backend API:** `http://localhost:5001`

Chapter 7: Privacy, Security Governance & Troubleshooting

100% Local Data Governance

Minomeet AI is engineered from the ground up for strict confidentiality:

- **Zero External Telemetry:** No Google Analytics, Mixpanel, or third-party trackers.
- **Local JSON Archive:** All meeting transcripts and summaries reside on your local machine at `server/data/db.json`.
- **Direct API Routing:** When using cloud AI, requests flow directly from your machine to the LLM API without intermediary proxy servers.

Frequently Asked Questions & Troubleshooting

Question / Issue	Solution & Best Practice
Microphone not capturing system audio	When recording video conferences, choose "Mixed (Mic + System)" and select the Chrome Tab or Entire Screen with "Share System Audio" checked.
"AI Model Not Configured" alert	Click the "Configure AI Model & API Key" banner on the Home page and enter your Google Gemini, OpenAI, Claude, or Groq API key in Settings.
Can I use Minomeet 100% offline?	Yes! Install Ollama locally (e.g. <code>ollama run llama3.3</code>) and select Ollama in Settings. Zero bytes leave your machine.

Minomeet AI — Autonomous Meeting Intelligence

Open Source • MIT Licensed • Created with ❤️ by [@shareefmx](#)